TELIPORT
Script Your Future
Season 4

TATA
TATA ELXSI

Case Study 2

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THINK. BUILD. CREATE. WIN

TELIPORT
Script Your Future Season 4

Test a battery before you build it

VOLT-FORGE connects your battery design decisions to what they physically change

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PROBLEM UNDERSTANDING

One change affects the whole battery

A Pi 5 gateway needs a **10-minute backup**. Fewer cells cost less, but each works harder. Wiring, cell variation and cooling change the answer together. We model one small cell group; ageing and packaging come later.

DESIRABILITY

Make the trade-off visible

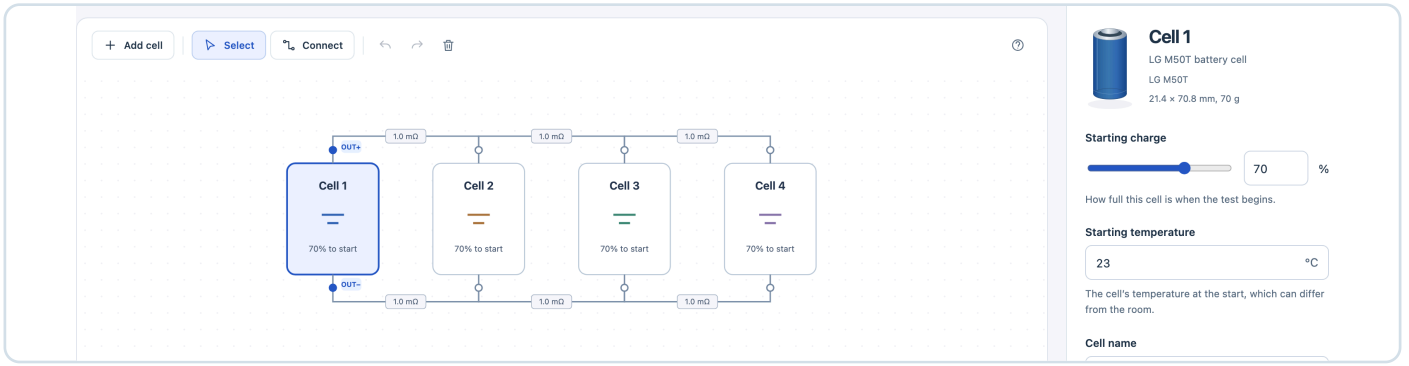
A student can change a circuit and explain what happens. An engineer can compare candidates before a bench test. Build, test, inspect the exact values and export the evidence, **in one workspace**.

TECHNICAL FEASIBILITY

How few cells can do this job?

Two cells passed 33/33 required simulations, including weak-cell cases; one cell was refused. The browser solves current and cell heating together. The table shows the trade-off against 3 and 4 cells. Board limits still need checking.

PROTOTYPE & EVIDENCE Test it out: **volt-forge-app.vercel.app**



Build it. Edit the actual circuit, its cells and the assumptions the solver uses.



Compare it. Same job, fewer cells: lower cell cost and mass, less electrical margin.

Parallel design	Required cases	Worst cell current	Worst cell temperature	Cell mass	Cell cost
4 cells	51/51	3.34 A	26.1 °C	280 g	₹4,824
3 cells	42/42	3.86 A	27.7 °C	210 g	₹3,618
2 cells	**33/33**	**5.49 A**	**31.8 °C**	**140 g**	**₹2,412**

Frozen study: same-end outputs; mismatch, thermal and converter sensitivities. Case counts vary by cell positions. Temperatures are simulated. Cost and mass cover cells only.

4,006 simulated cases.
13 design variants. Rechecked on the final engine.

54 published experiments.
194 files checksum-verified; 2 module records compared.

15/15 numerical checks.
Electrical and heat calculations checked against PyBaMM.

Assumptions, finance and sources: **volt-forge-app.vercel.app/appendix**

ENGINEERING QUALITY

Know what the evidence supports

Numerical checks passed. **All 12 comparisons with published measurements missed** voltage and temperature bands. Predictive accuracy remains unestablished. Monitoring and virtual cutoffs support investigation; physical board protection is unverified.

INNOVATION

Change it, then challenge it

Find the smallest design tests each size at normal conditions. Pressure tests show where it stops working. Every saved result keeps its inputs, model and software versions for repeatable comparisons.

BUSINESS VIABILITY

Start with teaching labs



41.3 lakh engineering students

15.6 lakh in three target disciplines (AISHE 2021–22)

130 labs at 5% assumed reach

Students free. Labs **₹50,000/year**; teams **₹5 lakh/year**. Proposed prices and adoption are untested.

25 W supply-rating scenario · 10 min · 23 °C · 70% charge · assumed 90% converter and 3.3 V floor · 8.42 A proxy